

BIJENDRA MANANDHAR

Senior Developer

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PROFESSIONAL SUMMARY

Accomplished Full-Stack Developer with 9+ years of experience engineering scalable web platforms and API-driven systems. Specializes in architecting interactive, self-paced e-learning platforms featuring modular course management, multimedia delivery, progress dashboards, and certificate generation. Recognized for reducing query latency, accelerating deployment cycles, and mentoring technical teams. Extensive track record delivering mission-critical systems for public institutions, humanitarian organizations, and international clients. Fluent in modern cloud infrastructure (AWS, GCP), real-time communication layers, and AI/LLM integrations. Based in Kathmandu with proven capability to contribute to climate-resilience and urban sustainability initiatives.

TECHNICAL SKILLS

Backend	Node.js · NestJS · Express.js · PHP Laravel · REST APIs · GraphQL · TypeScript
Frontend	Next.js · Vue.js · React · HTML5 Canvas · Responsive Web Design
Databases	PostgreSQL (stored procedures, triggers, partitions) · MongoDB · MySQL · Redis
Cloud & DevOps	AWS (S3, Lambda) · GCP · Docker · Nginx · Jenkins CI/CD
Integrations	Firebase · Bull Queue · RabbitMQ · WebSockets · Stripe · Google Drive API
AI / LLM	LLM API integrations · AI-powered feature research and delivery
GIS & Maps	Leaflet.js · Spatial data visualization · GIS map layers
Methods	Agile/Scrum · TDD · Database optimization · Technical mentorship

PROFESSIONAL EXPERIENCE

Senior Software Developer

2017 – Present

EK Solutions · Juwagal, Lalitpur

- Conceived and delivered an end-to-end interactive e-learning web platform (NestJS + Next.js) featuring hierarchical course catalogs, video/document lesson delivery, quiz scoring, learner dashboards, and automated certificate issuance — achieving a measurable increase in learner course-completion rates.
- Engineered real-time learner communication via WebSockets, enabling live Q&A sessions and instructor-to-learner notifications that reduced average response time by over 60%.
- Overhauled PostgreSQL schema using advanced indexing, table partitioning, stored procedures, and triggers — slashing critical query execution time by approximately 70%.
- Established CI/CD pipelines using Jenkins and Docker, compressing release cycles from bi-weekly to daily deployments and eliminating manual-deployment errors.
- Spearheaded asynchronous content-processing infrastructure with Bull Queue and RabbitMQ, enabling the platform to handle concurrent course uploads without service degradation.
- Orchestrated cloud media storage and distribution via AWS S3 and Lambda, supporting high-availability streaming for 500+ concurrent users.
- Completed an intern training program covering database fundamentals and Node.js; received a formal mentorship-excellence certificate for cultivating a high-performing junior cohort.
- Collaborated with cross-functional teams of 30+ to scope, prioritize, and ship multiple SaaS products within agreed timeframes and budgets.

Web Application Developer

2014 – 2016

Applied Innovation · Juwagal, Lalitpur

- Progressed from Junior Developer to Senior Web Application Developer within 24 months, demonstrating rapid skill acquisition and consistent delivery.
- Constructed loan-management web systems for Nepali financial institutions using Laravel and MySQL, processing thousands of records with high data integrity and audit compliance.
- Designed and integrated REST APIs facilitating seamless data exchange between third-party financial platforms and internal systems.

KEY PROJECTS

Interactive E-Learning Platform (*School*) · *NestJS · Next.js · PostgreSQL · AWS S3 · Redis · Firebase Auth*

- Architected a fully self-paced web LMS with structured course hierarchies, multimedia lesson delivery, interactive quizzes, progress-persistence dashboards.
- Implemented Redis-based session caching reducing authentication overhead by 55% and supporting 1,000+ concurrent learner sessions.

Disaster Management System — Nepal Red Cross Society (NRCS) · *mis.nrccs.org*

- Developed a mission-critical CMS for a national humanitarian organization integrating GIS map layers (Leaflet.js), real-time SMS alerts (Sparrow SMS), and MySQL-backed disaster-data visualization — directly aligned with climate-resilience and urban adaptation goals.

RECKI — Technical Lead

- Directed cloud infrastructure (GCP, AWS), established Jenkins CI/CD pipelines, and pioneered LLM/AI integrations into a consumer-facing rewards platform serving international users across multiple currencies.

IWI Myplace · *my-place.jp · Express.js · Vue.js · AWS · PostgreSQL*

- Crafted CMS modules and REST APIs with PostgreSQL stored procedures, Firebase push notifications, AWS S3 media hosting, and Google Drive API integration for a Japanese property-search platform.

RSP Nepal · *rspnepal.org · NestJS · Next.js · PostgreSQL · AWS*

- Delivered a high-performance CMS and API layer for a Nepali social enterprise, leveraging advanced PostgreSQL optimizations (table partitions, triggers, stored procedures) to sustain sub-100ms response times under peak load.

TTC · *uat-ttc.ekbana.info · NestJS · Next.js · Python · PostgreSQL · AWS · Rabbit MQ · Redis · BullQueue*

- TTC is a retail company undergoing full digital transformation, similar to Bhatbhateni. We're building all business modules from scratch, covering sales, HR, inventory, and internal companies under the TTC group.
- Additionally, integrating an AI-powered chat system to enhance user experience and operational efficiency. This includes intelligent assistance for internal users, automated customer support, real-time query handling, and AI-driven insights across modules using LLM-based services with token-based access control and scalable background processing.

EDUCATION

Bachelor of Computer Application

2023

Indira Gandhi National Open University

Intermediate +2 (Science)

2008

Kathmandu Institute of Science and Technology

CERTIFICATIONS & ACHIEVEMENTS

Intern Mentorship Excellence Certificate

EK Solutions

Formally recognized for cultivating a high-performing junior developer cohort through structured training in database fundamentals and Node.js development.

LANGUAGES

English · Nepali · Newari

REFERENCE

Pratik Tuladar Senior Project Manager, EKbana
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